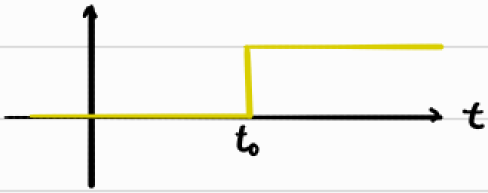
Unit step function [Heavyside function]

$$u(t-t_0)\begin{cases}0 & \text{if } t<t_0\\ 1 & \text{if } t>t_0\end{cases}$$



Now we take the Laplace of the heaviside function,

$$\Rightarrow \mathcal{L}\{u(t-t_0)\} = \int_0^\infty u(t-t_0)e^{-st}\,dt$$
$$= \int_0^{t_0} 0\cdot e^{-st}\,dt + \int_{t_0}^\infty 1\cdot e^{-st}\,dt$$
$$= \int_{t_0}^\infty e^{-st}\,dt$$
$$= \int_0^\infty e^{-s(\hat{t}-t_0)}\,d\hat{t}$$
$$= e^{-t_0 s}\int_0^\infty e^{-s\hat{t}}\,d\hat{t}$$
$$= e^{-t_0 s}\,\mathcal{L}\{1\}$$
$$\boxed{\mathcal{L}\{u(t-t_0)\} = e^{-t_0 s}\left(\tfrac{1}{s}\right)}$$

Let $\hat{t} = t - t_0$

$t = \hat{t} + t_0$ — constant

$dt = d\hat{t}$

When $t = t_0 \Rightarrow \hat{t} = 0$

$t = \infty \Rightarrow \hat{t} = \infty$

Consider $\mathcal{L}\{g(t-t_0)\,u(t-t_0)\}$,

$$\mathcal{L}\{g(t-t_0)\,u(t-t_0)\,e^{-st}\} = \int_0^\infty g(t-t_0)\,u(t-t_0)\,e^{-st}\,dt$$
$$= \int_0^{t_0} 0\cdot g(t-t_0)e^{-st}\,dt + \int_{t_0}^\infty 1\cdot g(t-t_0)\,e^{-st}\,dt$$
$$= \int_0^\infty g(\hat{t})\,e^{-s(\hat{t}-t_0)}\,d\hat{t}$$
$$= e^{-st_0}\int_0^\infty g(\hat{t})\,e^{-s\hat{t}}\,d\hat{t}$$
$$= e^{-st_0}\,\mathcal{L}\{g\}$$
$$\boxed{\mathcal{L}\{g(t-t_0)\,u(t-t_0)\} = e^{-st_0}\,G(s)}$$

Terminology

$g(t)$ ~ Unshifted Base Func.

$G(s)$ ~ Laplace Transfrom of $g(t)$

$g(t-t_0)$ ~ Shifted Base Func.

Example 1

$g(t-2) = e^{(t-2)} \Rightarrow g(t) = e^t$

$$\mathcal{L}\{u(t-2)\,e^{(t-2)}\} = e^{-s(2)}\,\mathcal{L}\{e^t\}$$
$$= e^{-2s}\,\frac{1}{s-1}$$

Example 2

$g(t-3) = t \Rightarrow g(t) = t+3$

$$\mathcal{L}\{u(t-3)\cdot t\} = e^{-s(3)}\,\mathcal{L}\{t+3\}$$
$$= e^{-3s}\left[\frac{1}{s^2}+\frac{3}{s}\right]$$ #

Inverting t-shift

Example 1

$$\mathcal{L}^{-1}\left\{ e^{-5s} \frac{2}{s^3} \right\}$$

$t_0 = 5$ $\quad G(s)$

1) Find unshifted base function, $g(t)$

$$g(t) = \mathcal{L}^{-1}\{G(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\} = t^2$$

$\therefore g(t) = t^2$

2) Sub into $u(t-t_0)\, g(t-t_0)$

$$\Rightarrow \mathcal{L}^{-1}\left\{ e^{-5s} \frac{2}{s^3} \right\} = u(t-5)\, g(t-5)$$

$$= u(t-5)(t-5)^2 \ \#$$

Example 2

$$\mathcal{L}^{-1}\left\{ e^{-8s} \frac{4}{s^2+16} \right\}$$

$t_0 = 8$ $\quad G(s)$

1) Find unshifted base function

$$g(t) = \mathcal{L}^{-1}\left\{\frac{4}{s^2+4^2}\right\}$$

$$= \sin(4t)$$

2) Sub into $u(t-t_0)\, g(t-t_0)$

$$\Rightarrow \mathcal{L}^{-1}\left\{ e^{-8s} \frac{4}{s^2+16} \right\} = u(t-8)\, g(t-8)$$

$$= u(t-8) \sin(4(t-8))$$

$$= u(t-8) \sin(4t-32)$$

Delta Function, $\delta(t)$

The 2 properties are ...

① Infinite height, zero width

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$

② Area = 1

trivially small num.

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1 \quad \text{OR} \quad \int_{t_0-\varepsilon}^{t_0+\varepsilon} \delta(t-t_0)\, dt = 1$$

[Area of entire graph is 1] [Area of spike only is 1]

Example 1

$$\int_0^{10} e^t \delta(t-2)\, dt = \int_{2-\varepsilon}^{2+\varepsilon} e^t \delta(t-2)\, dt$$

$$= e^2 \int_{2-\varepsilon}^{2+\varepsilon} \delta(t-2)\, dt$$

Since $\delta(t-2)$ is zero everywhere except $t=2$, $e^t = e^2$

$$= e^2 \cdot 1$$

$$= e^2 \ \#$$

Consider $\mathcal{L}\{\delta(t-t_0)\, g(t)\}$

$$\mathcal{L}\{\delta(t-t_0)\, g(t)\} = \int_0^{\infty} \delta(t-t_0)\, g(t)\, e^{-st}\, dt$$

$$= \int_{t_0-\varepsilon}^{t_0+\varepsilon} g(t)\, e^{-st}\, \delta(t-t_0)\, dt, \quad t_0 > 0$$

$$= g(t_0)\, e^{-st_0} \int_{t_0-\varepsilon}^{t_0+\varepsilon} \delta(t-t_0)\, dt$$

$$\boxed{\mathcal{L}\{\delta(t-t_0)\, g(t)\} = g(t_0)\, e^{-st_0}} \ \#$$

Periodic functions

A function $y = f(x)$ is periodic if $f(x+N) = f(x)$, for all x

The smallest positive N is the period, P

$\Rightarrow \; f(x + \underbrace{P}_{\text{Period}}) = f(x)$, for all x

OR

$\Rightarrow \; f(x + 2\underbrace{L}_{\text{Half-period}}) = f(x)$, for all x

Anti-symmetrical

Odd function : $f(x) = -f(-x)$ [Reflect y-axis, then x-axis]

symmetrical

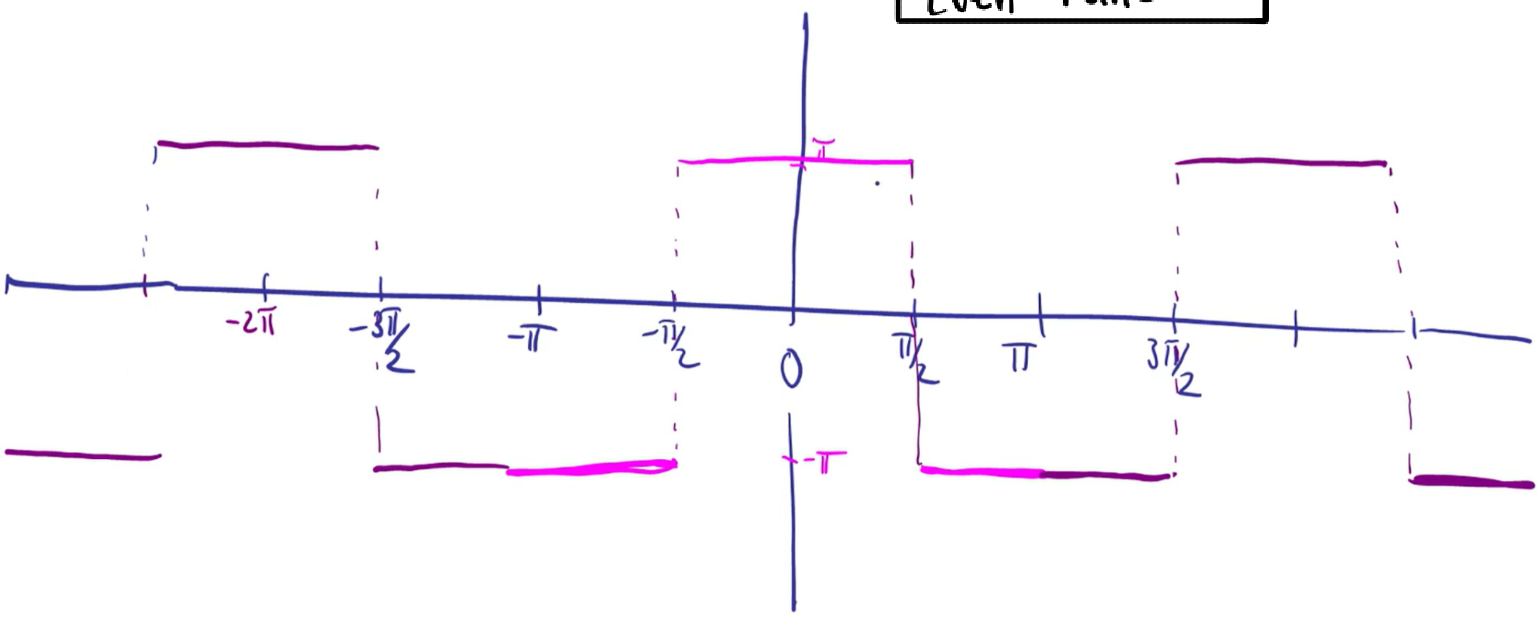
Even function : $f(x) = f(-x)$ [Reflect y-axis]

Example 4.15, step function

Define the periodic function, a *step function*,

$$f(x) = \begin{cases} -\pi & \text{for } -\pi < x \leqslant -\frac{\pi}{2} \\ \pi & \text{for } -\frac{\pi}{2} < x \leqslant \frac{\pi}{2} \\ -\pi & \text{for } \frac{\pi}{2} < x \leqslant \pi \end{cases} \quad \text{and } f(x) = f(x + 2\pi)$$

$\boxed{\text{Even Function}}$



Fourier Series

The Fourier Series of a 2π-periodic function f(x) can be written as

$$\boxed{f(x) = \underbrace{a_0}_{\text{y-offset of the shape}} + \underbrace{\sum_{n=1}^{\infty}\left[a_n \cos(nx) + b_n \sin(nx)\right]}_{\text{Adjust the shape}}}$$

$$\boxed{a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx}$$

This literally gives the avg value of f(x) $\left[\frac{\text{Total Area}}{2\pi}\right]$

Finds the "vertical centre" of the shape

$$\boxed{a_n = \underbrace{\frac{1}{\pi}}_{\text{scaling factor}}\int_{-\pi}^{\pi} \underbrace{\overbrace{f(x)}^{\text{Actual wave}}\,\overbrace{\cos(nx)}^{\text{test wave}}}_{\text{Resonance}}\,dx}$$

$$\boxed{b_n = \overbrace{\frac{1}{\pi}}^{\text{scaling factor}}\int_{-\pi}^{\pi} \underbrace{\overbrace{f(x)}^{\text{Actual wave}}\,\overbrace{\sin(nx)}^{\text{Test wave}}}_{\text{Resonance}}\,dx}$$

When test wave = Actual wave
$\int_{-\pi}^{\pi} A\cos^2(nx)\,dx = A\pi$

When test wave = Actual wave
$\int_{-\pi}^{\pi} A\sin^2(nx)\,dx = A\pi$

This new "resonance" function will only have a non-zero area when the waves sync up so that even their -ve areas become +ve (because (-ve × -ve = +ve)). If the waves don't sync, it will create equal amounts of +ve & -ve area that perfectly cancel out to zero

TIPS

Since our bounds are -π to π [Perfectly centered around y-axis],

$\int_{-\pi}^{\pi}(\text{Odd function}) = 0$

$\int_{-\pi}^{\pi}(\text{Even function}) = 2\int_{0}^{\pi}(\text{Even Function})$

Note that ... (Odd)(Odd) = Even

(Odd)(Even) = Odd

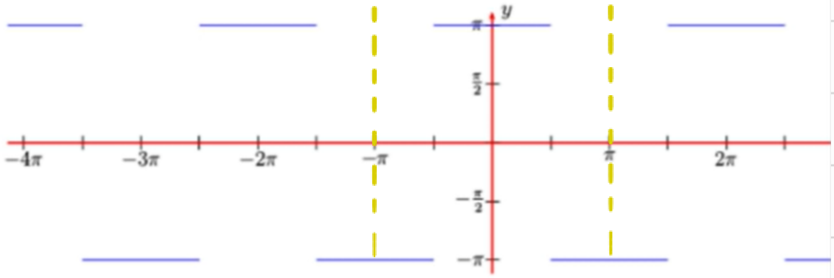
(Even)(Even) = Even

Example 4.2.1

Determine the Fourier series for the step function

$$f(x)=\begin{cases}-\pi & \text{for } -\pi<x<-\frac{\pi}{2}\\ \pi & \text{for } -\frac{\pi}{2}<x<\frac{\pi}{2}\\ -\pi & \text{for } \frac{\pi}{2}<x<\pi\end{cases} \quad \text{and } f(x)=f(x+2\pi)$$

$\boxed{\text{Even function}}$



$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{2\pi}\int_{-\pi}^{-\pi/2}(-\pi)\,dx + \int_{-\pi/2}^{\pi/2}\pi\,dx + \int_{\pi/2}^{\pi}-\pi\,dx$$

$\vdots$

$a_0 = 0$ #

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{\pi}(-\pi)\cos(nx)\,dx + \int_{-\pi/2}^{\pi/2}\pi\cos(nx)\,dx + \int_{\pi/2}^{\pi}(-\pi)\cos(nx)\,dx\right\}$$

$\vdots$

$$= \frac{1}{n}\left\{-\left[-\sin\left(\frac{n\pi}{2}\right) - \cancelto{0}{\sin(-n\pi)}\right] + \left[\sin\left(\frac{n\pi}{2}\right) + \sin\left(+\frac{n\pi}{2}\right)\right] - \left[\cancelto{0}{\sin(n\pi)} - \sin\left(\frac{n\pi}{2}\right)\right]\right\}$$

$$= \frac{4}{n}\sin\left(\frac{n\pi}{2}\right)$$

$$a_n = \begin{cases}\frac{4}{n}, & n=1,5,9\ldots\\ 0, & n=2,6,10\ldots\\ -\frac{4}{n}, & n=3,7,11\ldots\\ 0, & n=4,8,12\ldots\end{cases}$$ #

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{\pi}(-\pi)\sin(nx)\,dx + \int_{-\pi/2}^{\pi/2}\pi\sin(nx)\,dx + \int_{\pi/2}^{\pi}(-\pi)\sin(nx)\,dx\right\}$$

$\vdots$

$b_n = 0$ #

$\therefore$ Since $f(x)$ is even, $\sin(nx)$ is odd, $\int_{-\pi}^{\pi}(\text{even})(\text{odd}) = \int_{-\pi}^{\pi}(\text{odd}) = 0$

$\therefore$ We didn't even need to calc. this.

$$\therefore \text{Fourier series} = 0 + \sum_{n=1}^{\infty}\left\{\left[\frac{4}{n}\sin\left(\frac{n\pi}{2}\right)\right]\cos(nx) + 0\right\}$$

$$= \sum_{n=1}^{\infty}\frac{4}{n}\sin\left(\frac{n\pi}{2}\right)\cos(nx)$$

$$= \underbrace{4\cos(x)}_{n=1} - \underbrace{\frac{4}{3}\cos(3x)}_{n=3} + \underbrace{\frac{4}{5}\cos(5x)}_{n=5} + \ldots$$ #